The Price of Protecting Small Firms: Evidence from Public Procurement*

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Abstract

Governments restrict public tenders to small firms, yet the fiscal cost is unknown. Exploiting a natural experiment in Sao Paulo, Brazil—where medical supplies were uniquely exempt from mandatory SME-only tenders until a 2018 legal reinterpretation ended this exemption—I estimate these costs using a difference-in-differences in reverse design. Open tenders yielded prices 12–13% lower and attracted 19% more firms. A Gelbach decomposition reveals that increased participation explains only 6% of the price reduction; most savings come from fiercer bidding. The fiscal cost for one product group alone is R$84.5 million over 18 months.

Keywords: public procurement, SMEs, policy costs, restricted public tenders.

JEL No.: H32, H57, L26, L53, O12, R12.

1. Introduction

Governments worldwide use public procurement to promote small and medium-sized enterprises (SMEs), directing a substantial share of public spending through set-aside programs and bid preferences (Thai, 2017; Bosio et al., 2022). The scale of these policies is vast: in the European Union, SMEs account for nearly 99.8% of all registered firms and receive a large share of public contract value (PwC, 2014); in Brazil, where approximately

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97% of firms are SMEs representing half of formal employment (Bastos et al., 2018), federal law mandates exclusive SME tenders for all items valued below R$80,000. The rationale is well established—SMEs are widely regarded as engines of job creation, local development, and innovation (SIGMA, 2016)—and there is evidence that government contracts can catalyze firm growth (Mel et al., 2008; Freedman, 2013), though the effectiveness of this channel depends on the quality of management practices (McKenzie and Woodruff, 2015) and how capital is provided (Fafchamps et al., 2014). Yet despite the global prevalence and fiscal magnitude of these policies, credible evidence on what they *cost* the public sector is remarkably scarce.

There are strong theoretical reasons to expect these costs to be large. The auction literature provides a sharp prediction: Bulow and Klemperer (1996) show that attracting one additional bidder generates greater surplus for the auctioneer than any mechanism design refinement, implying that policies restricting participation impose a first-order efficiency loss. Krasnokutskaya and Seim (2011) formalize this in the procurement context, showing that bid preferences reduce both the number of bidders (extensive margin) and the aggressiveness of their bids (intensive margin). Empirically, Marion (2007) estimates a 3.8% cost increase from small business preferences in California highway procurement, and Athey et al. (2013) document similar trade-offs in US timber auctions. But outside the US, credible estimates of set-aside costs are virtually absent—and in developing economies, where SME policies are most prevalent, they are nonexistent (Nakabayashi, 2013; Szucs, 2024; Decarolis et al., 2024).

This asymmetry matters: while a growing literature documents the benefits of SME procurement policies—firm growth (Ferraz et al., 2016), employment (Szerman, 2023), supplier network expansion (Cardoza et al., 2016)—without credible cost estimates, policymakers cannot evaluate the efficiency-equity trade-off that these policies entail. Recent advances in causal inference (Athey et al., 2019; Rambachan and Roth, 2023)

offer new tools for this evaluation, yet their application to procurement set-asides remains nascent.

This paper provides new evidence on this question by exploiting a quasi-experimental variation in Sao Paulo, Brazil, where the SME set-aside policy operates through a distinctive institutional mechanism: *opt-out costs*. Public buyer units (PBUs) that wish to conduct open tenders rather than SME-only tenders must justify their choice through a costly formal process, subject to scrutiny by audit courts and potential sanctions. These opt-out costs effectively penalize the efficient procurement choice, shifting the default from open competition to restricted tenders. I exploit the timing of a policy change (March 2018) that extended these opt-out costs to a previously exempt product group (Group 65, medical and hospital supplies), using time and cross-sectional variation to estimate the fiscal cost of this institutional design.

The institutional setting provides unusually clean identification. Between August 2014 and February 2018, the state of Sao Paulo exempted Group 65 (medical and hospital supplies) from the mandatory SME-only tender requirement that applied to all other product groups, based on a joint agreement between the state government and the audit court that health items constituted strategic goods warranting open competition. In March 2018, a legal opinion from a different oversight agency reversed this interpretation, subjecting Group 65 to the same SME-favoring rules as all other groups. This policy change was driven by a legalistic reinterpretation of isonomy principles rather than by changes in procurement outcomes for Group 65, making it plausibly exogenous to the outcomes studied.

However, examining how this institutional change occurred allows the costs of the SME policy to be assessed only indirectly. Instead of using a standard difference-in-differences (DiD) method, I use a variation of this method known as difference-in-differences in reverse (DiDiR), or 'time-reversed DiD' (Kim and Lee, 2019). In the DiD method, there

is a control group that is never treated and a treatment group that is treated at some point in time. In the DiDiR method I employ, the control group is always treated (instead of always untreated), and the other group is the 'switched group,' subject to the change in policy.

In Sao Paulo, the government can execute public tenders for SMEs only. Between August 2014 and February 2018, the government's default choice consisted of executing SME-only tenders for all items with a value less than or equal to R$80,000, except for a group of items identified as code 65, which was only allowed to execute open tenders in this period.

PBUs can avoid restricted tenders by justifying the choice to audit courts—but this costly process constitutes the opt-out costs that discourage open competition. From March 2018 on, after a change in the law's interpretation, Group 65 became subject to these opt-out costs just as any other group of purchased items. Thus, the 'always-treated' group here consists of all groups of items but Group 65, comprising the switched group.

Notably, DiDiR identifies pre-switch-period effects; i.e., it estimates effects for the past (Kim and Lee, 2019), assessing the costs of the policy switch indirectly. I estimate this pre-switch-period effect on the switched group, comparing the observed outcomes for Group 65 before the shift in policy and the outcomes that would have occurred for this group if there had been opt-out costs before March 2018.

The DiDiR method reveals that the negotiated prices are, on average, between 12% and 13% lower for Group 65 than other groups before March 2018 (13–14 log points across specifications). Moreover, in the pre-period, the number of participants in Group 65 is approximately 19–20% higher than that in other groups of items in the short-term window. The number of valid bids follows the same pattern: there are approximately 16–17% more valid bids in Group 65 than in the other groups of items. Finally, before the policy change, sellers that won tenders for items in Group 65 were approximately

5–11 km further away from public buyer units (PBUs), depending on the specification. These price effects are concentrated among high-value items, where the pool of capable non-SME suppliers is broader and the efficiency gains from open competition are larger. These are nominal estimates; IPCA-deflated coefficients range from 7% to 12% (see Section 4.5). Translating the nominal estimates into monetary terms, the fiscal cost of the SME-only restriction for Group 65 alone amounts to R$84.5–85.8 million (approximately US$17 million) over the 18-month pre-period—about 12% of this product group's total procurement value. Since Group 65 represents only 27% of total procurement on the platform, a conservative extrapolation places the aggregate fiscal cost of SME restrictions at approximately R$310 million (∼US$62 million) over the same period—resources that could fund alternative SME promotion instruments at a fraction of the efficiency loss.

The results are supported by a comprehensive set of robustness checks. Placebo tests using fake treatment dates confirm that the estimated effects are not driven by pre-existing differential trends. Alternative clustering strategies and winsorization of outcomes corroborate the main findings. Extensions to IPCA-deflated real prices, procurement efficiency ratios, and winner composition analysis provide evidence on the channels through which the policy operates. Advanced econometric methods in the Appendix provide further corroboration: Lee bounds demonstrate that the price effect survives correction for sample selection, quantile DiD reveals that the effect is concentrated at lower quantiles of the price distribution, and a causal forest analysis identifies item quantity as the key moderator of treatment effect heterogeneity.

This paper contributes to two strands of the literature. First, it adds to the growing body of evidence on the costs of restricting competition in public procurement (Bosio et al., 2022; Szucs, 2024; Decarolis et al., 2024; Bandiera et al., 2009; Best et al., 2023), providing estimates from a large developing economy where SME policies are especially prevalent. Second, it contributes to the literature on set-asides and bid preferences

(Athey et al., 2013; Krasnokutskaya and Seim, 2011; Marion, 2007) by documenting how these costs vary across item types, offering a richer picture of how procurement efficiency losses distribute across heterogeneous goods. A distinguishing feature of this paper is its methodological breadth: beyond the standard reduced-form analysis, I deploy a suite of advanced methods—including parallel trends sensitivity analysis, sample selection bounds, causal forests, quantile regressions, and mechanism decomposition—to stress-test the findings from multiple angles.

The remainder of this paper is organized as follows. Section 2 provides a brief overview of the institutional background related to SME law and public procurement in Sao Paulo, Brazil. Section 3 describes the datasets and sample definitions. Section 4 presents the empirical analysis, including heterogeneity analysis, robustness checks, and a fiscal cost quantification. Section 5 concludes with policy implications.

2. Institutional Background

This section provides a brief institutional background on SMEs' participation in the context of public tenders in Sao Paulo, Brazil. I focus on the details that are most pertinent for the empirical analysis.

First, I briefly discuss general aspects of public procurement and the legislation on micro and small businesses in Brazil, highlighting the established criteria for the occurrence of SME-only public tender. Second, I describe how the state of Sao Paulo applied this law in accordance with its own understanding of exclusive tenders for health items.

2.1. Public Procurement and SME Law in Brazil

Public procurement constitutes a relevant part of economies worldwide (Bosio et al., 2022). In 2016, OECD countries spent an average of approximately 12% of GDP on public

procurement, while in Brazil, this proportion was approximately 10% in the same year. Recent evidence from Brazil shows that procurement institutions and their enforcement have significant effects on firm dynamics and employment (Szerman, 2023; Colonnelli and Prem, 2022).

As in many other countries, Brazilian law establishes as a general rule that all purchases, services, and works hired by the public administration should be subject to a public tender. Federal Law 8,666/1993 institutes a general framework applicable to all public bids in the country, and all three government branches must adhere to this framework.

Entities directly or indirectly controlled by the federal, state, or municipal governments must comply with the government procurement rules. Federal, state, and municipal governments, autonomous government entities, public foundations, regulatory agencies, state-owned companies, and mixed capital companies controlled by the government are subject to these rules. These entities are known as public buyer units (PBUs).

Although the public administration may decide to make purchases centrally, in Brazil, almost all acquisitions are decentralized and made by PBUs. A ministry or bureau may consist of many PBUs that have budgetary autonomy and make purchases from private companies. PBUs may contract a wide variety of products and services from private companies, including engineering and infrastructure work. However, this paper focuses on analyzing the acquisition of common and standardized goods.

The primary purpose of a bidding process conducted by a PBU is to seek the best contract possible for the government. The Brazilian public procurement law provides guidelines on how the procurement process should be organized and executed. In some cases, public tenders for SMEs are subject to different treatment.

The Brazilian federal SME law was enacted in 2006 to regulate favored, simplified and differentiated treatment for this sector, as provided for in the Federal Constitution.

This law's explicit goal was to promote SMEs' economic and social development and competitiveness as a strategy for job creation, income distribution, social inclusion, reduced informality, and a strengthened economy.

The Brazilian SME law adopts the following classification for companies, according to their annual gross revenue: (i) microbusiness: annual gross revenue of R$360,000 or less (roughly US$72,000); and (ii) small business: annual gross revenue greater than R$360,000 and less than or equal to R$4,800,000 (between US$72,000 and US$960,000). In this paper, these companies are referred to as SMEs.

SMEs enjoy many benefits provided by law, including tax benefits and fewer bureaucratic requirements to adhere to. In addition, public tenders held at the federal, state, and municipal levels can grant differentiated and privileged treatment to SMEs to promote economic and social development, increase public policies' efficiency, and stimulate technological innovation.

The content of the SME law, in its 2006 version, indicated that the public administration *could* create tenders exclusively for the participation of SMEs in purchases in which the item value was up to R$80,000 (approximately US$16,000). Thus, choosing tenders for SMEs only was optional for PBUs.

However, the federal SME law underwent a significant change from SMEs' exclusivity in tenders in 2014. The term "could" was replaced with "must," making it mandatory to execute exclusive public tenders for SMEs up to a value per item of R$80,000.

The law changed PBUs' default choice if the item value fell below the threshold of eighty thousand reais: previously, the default choice was open bids. As of 2014, the standard option for PBUs is to execute public tenders for SMEs only, unless the bid's conditions fall within the exceptions provided for in the updated legislation.

PBUs can avoid restricted bids if at least one of the following conditions is met: (i) there are two or fewer potential competing SME suppliers that are locally or regionally

based and able to comply with the notice requirements; or (ii) PBUs consider that the differentiated and simplified treatment for SMEs might not be advantageous for the public administration. Thus, PBUs choose whether the public tender is restricted, but they must justify their choices to their watchdogs, such as audit courts or the judiciary.

On the one hand, this discretion provided for by law can be beneficial since PBUs can more efficiently choose the bidding type to be carried out. However, there are costs involved in the process of avoiding bids restricted to SMEs. For each bidding procedure, PBUs must create an extensive report listing in detail the reasons that justify the use of an open bidding process to the detriment of a bidding process restricted to SMEs.

Additionally, these PBU justifications are subject to scrutiny by both the audit courts and the judiciary. If these bodies consider the arguments unfounded or insufficient, administrative proceedings and punishments may be brought against the public agents responsible for planning and executing the bid in question. Thus, this discretion brings costs to PBUs. I call these costs associated with avoiding SME-only tenders *opt-out costs*.

2.2. SME-only Public Tenders: Group 65 As an Exception

Sao Paulo is the wealthiest and most populous state in Brazil. This state accounts for approximately 23% of the total population and nearly one-third of Brazil's GDP (nearly US$500 billion in 2018). This amount is equivalent to the GDP of countries such as Sweden, Poland, and Belgium and more than twice the GDP of Portugal, Greece, and Finland. The state of Sao Paulo has a diversified economy driven by the automobile, textile, chemical, aeronautical, and computer industries, in addition to services such as finance and agriculture.

Since 2005, all PBUs in the state of Sao Paulo have been required to purchase common goods and services through Bolsa Eletrônica de Compras (BEC), an electronic purchasing platform. BEC operates as a centralized electronic reverse auction system (*pregão eletrônico*), in which registered suppliers submit progressively lower bids in real

time until the auction closes, facilitating transparent price competition. This design is comparable to other electronic procurement platforms adopted worldwide, such as Chile's ChileCompra and South Korea's KONEPS, which similarly aim to reduce transaction costs, increase supplier participation, and improve price outcomes through digitized and standardized processes. The BEC figures are revealing: in 2019, approximately R$13 billion (about US$3 billion) in trade was conducted on this e-platform. Since its implementation in 2005, BEC has moved more than R$105 billion (about US$20 billion) in negotiations; 860,000 purchase offers have been made, and approximately 4.8 million items have been sold.

Despite being subject to federal laws, Brazilian states have the prerogative to regulate or interpret these laws' specific elements. The state of Sao Paulo, for example, has a specific interpretation of how to apply SME law in public procurement.

Since BEC's implementation, the Sao Paulo state government has considered the group of items consisting of health-related products, including medication and hospital supplies (code 65), as a strategic set of items in the public procurement process. For example, bidding procedures related to medication can be carried out only with dynamic reverse auctions (*pregão*) or sealed bidding (*convite*). Direct negotiation (*dispensa de licitação*), which is very common in the purchase of other types of common goods, has always been prohibited for Group 65 in Sao Paulo.

Between 2006 and 2014, when the first version of the SME law was enforced, PBUs located in the state of Sao Paulo had the default choice to hold open tenders; it was optional to set procedures restricted to SMEs. During this period, in accordance with this legal arrangement, there was low adherence to restricted bids; items in these bids accounted for 6–13% of the total number of items bid for in the state of Sao Paulo. However, for Group 65, there were no bids exclusively for SMEs in this same period. The government had an internal orientation to hold open tenders for Group 65, with the

explicit agreement of the audit court of the state of Sao Paulo (TCE-SP).

After 2014, with the update of the federal law on SMEs, the default choice was to execute SME-only tenders if the item value was less than or equal to R$80,000. In the state of Sao Paulo, if PBUs consider that any item in a bidding process falls within the exceptions provided for by law, they must justify in detail the reasons for the non-execution of an exclusive bid for SMEs through a report sent to TCE-SP.

This justification offered by PBUs to avoid restricted tenders not only requires excessive work effort for PBUs but also is subject to the scrutiny of the TCE-SP and the judiciary. Public officers can face punishment if there are irregularities or failure to comply strictly with the law.

With the introduction of considerable opt-out costs for PBUs, this incentive scheme appears to be effective in encouraging the choice of offering exclusive bids to SMEs: adherence to this procedure has increased from approximately 13% to almost 70%, on average, in subsequent periods.

However, between August 2014 and February 2018, bids related to Group 65 did not change. The Sao Paulo state government and TCE-SP, in a joint agreement, used a specific interpretation of the new version of the federal SME law to leave Group 65 as an exception. The federal law provides that the requirement of restricted bids to SMEs does not apply when "it is not advantageous to the public administration or represents a loss to public resources." Thus, using this guideline, which allows for a high degree of discretion, the state of Sao Paulo and TCE-SP operated on the interpretation that because Group 65 constitutes a set of strategic items to meet such an essential public policy, it should always be subject to open bids. Thus, in this period, there was no bidding restricted to SMEs involving Group 65.

However, in November 2017, a different control agency in the state of Sao Paulo, PGE-SP, issued a document containing a legal opinion that changed this interpretation.

This document reinforces that the principle of isonomy in law enforcement should prevail in public procurement. Therefore, items in Group 65 should be considered for public procurement purposes in the same way as other groups.

Hence, as of March 2018, the opt-out costs also apply to Group 65. Between March 2018 and December 2019, adherence to exclusive bids for SMEs for this group was, on average, 43%. This lower proportion of restricted bids for SMEs for Group 65 compared with other groups may be due to the existence of more oligopolies in this group of items, which include, for example, medication. Thus, in many cases, it is not possible to find at least three potential suppliers that are SMEs.

Two features of this institutional change are particularly important for the empirical strategy. First, the trigger for the policy change was a legalistic reinterpretation of the isonomy principle by PGE-SP, not a response to deteriorating procurement outcomes for Group 65. There is no evidence that the decision to extend SME-only tender rules to Group 65 was motivated by price levels, competition patterns, or supplier complaints in this specific product group. Second, the timing of the change was driven by bureaucratic processes within PGE-SP and its communication to PBUs, making it plausibly exogenous to the procurement outcomes studied in this paper. These features provide the institutional basis for the identifying assumption that, absent the policy change, Group 65 and other groups would have followed parallel outcome trajectories in the post-period.

3. Data

This section describes the data source and details the sample characteristics used in the empirical section.

I use administrative data on bidding-level public procurement tenders of common goods and services in the state of Sao Paulo, Brazil, from January 2016 to December 2019.

All transactions took place on the electronic procurement platform called BEC, available for all PBUs across the state. The SEFAZ/SP (State Finance Secretariat) is responsible for the operational management and centralization of BEC's bidding data. An important advantage of this data source is that BEC records the universe of standardized goods procurement in the state, eliminating selection concerns that arise in settings where data coverage is partial or voluntary.

On BEC, 1,344 PBUs regularly make purchases. These entities include state-level bureaus from the executive, legislative, and judiciary branches in the state of Sao Paulo as well as other affiliated entities, such as municipalities located in the state of Sao Paulo and private organizations. PBUs purchased 82,569 different items (goods) totaling 832,984 successful transactions from 2016 to 2019.

BEC has a very detailed catalog of standardized goods and services organized in three levels of detail: group, class, and item. For instance, health items are classified as Group 65 (medical, dental, and hospital equipment and supplies). The item coded as 110639 refers to the drug 'Furosemide 40 milligrams, coated tablets, units', belonging to class 6531 (Medicines prescribed with or without ANVISA notification/registration) and Group 65.

Data are organized by purchase offer (PO), the electronic document issued by the PBU that identifies and quantifies the goods and services that will be purchased. A PO is defined by a 22-character code and may contain one or more items listed, but each item has its own purchase process. Thus, the purchase of an item is uniquely identified by the combination of the PO and the purchased item codes (POI).

There is a crucial variable for the empirical section, defined from the item group codes. It is a binary variable that assumes the value of 1 if the item belongs to Group 65 and 0 otherwise. The items in Group 65 constitute the 'switched group', i.e., the set of items affected by the purchasing policy change. Group 65 accounted for almost 27% of

all purchases from 2016 to 2019. All other groups of items comprise the 'control group.'

There are 75 groups of items, excluding Group 65. Between 2016 and 2019, the most significant groups were groups 89 (food products), 75 (office supplies), 86 (computer products), and 79 (cleaning materials), representing 37.5% of the total purchases in this period. The diversity of the always-treated group—spanning food, office supplies, technology, and cleaning materials—mitigates the concern that the comparison group is idiosyncratic or dominated by a single product category.

For each POI, there is information about item quantities, bid prices (winners and losers), the number of participant firms, the number of bids, whether the public tender was successful or not, the identification of the PBU, and firms' and PBUs' location, among other variables.

The empirical analysis focuses on four main outcome variables. First, the *negotiated price* (in logs) captures the final price at which the item was transacted; this variable is available only for completed items. Second, the *number of participant firms* (in logs) measures how many distinct firms submitted at least one bid, regardless of whether the item was ultimately purchased. Third, the *number of valid bids* (in logs) counts the total number of qualifying bids received per item. Fourth, *distance* measures the geographic distance in kilometers between the winning firm's location and the PBU. The regressions control for the log quantity ordered, the type of tender (sealed bid vs. reverse auction), and include item-level fixed effects that absorb all time-invariant heterogeneity across items—including differences in markets, pricing norms, and typical supplier pools.

To ensure that the estimates are not driven by compositional changes in the sample, the analysis restricts attention to items that appear in both the pre- and post-periods within each time window. The three time windows (6, 12, and 18 months around the March 2018 cutoff) provide a check on the stability of the estimates as the sample expands. The price and distance regressions are estimated on the subsample of completed items

(where a winning bid exists), while the participation and bid regressions use all items, including those where no firm was selected. An important implication of conditioning on completion is that the price and distance estimates may be affected by sample selection if the treatment itself influences whether items are successfully transacted; the Lee bounds analysis in the Appendix directly addresses this concern. Additionally, the breadth of item groups in the always-treated category—spanning food products, office supplies, computer equipment, cleaning materials, and dozens of other categories—provides substantial cross-sectional variation that strengthens the external validity of the estimates beyond any single product market.

Table 1 presents descriptive statistics for the key variables used in the empirical analysis, disaggregated by group (Group 65 vs. Others) and period (Pre vs. Post the policy shift in March 2018).

Table 1: Descriptive Statistics (18-month window)

	Group 65, Pre		Group 65, Post		Others, Pre		Others, Post	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Price (levels)	1,402.09	20,563.54	1,507.92	25,872.31	1,465.85	50,385.93	1,291.37	68,042.55
Log price	1.72	2.84	1.91	2.81	2.85	2.12	2.78	2.07
Number of firms	3.17	2.52	3.02	2.40	4.66	3.27	4.69	3.28
Log number of firms	0.88	0.73	0.84	0.71	1.30	0.72	1.30	0.72
Number of valid bids	11.10	15.49	10.99	16.51	10.13	15.74	10.47	16.87
Log number of valid bids	1.64	1.24	1.61	1.23	1.69	1.06	1.70	1.08
Distance (km)	238.54	270.42	215.08	239.63	164.78	182.23	170.28	186.07
N (completed / all)	65,606	77,206	66,822	85,181	246,277	287,500	271,011	323,836

Notes: Price and distance statistics computed on completed items (status = 1). Firm and bid statistics computed on all items.

4. Empirical Strategy and Results

4.1. Main Specification and Identification Strategy

The identification strategy exploits the timing of the policy change (March 2018) that affected only Group 65, using time and cross-sectional variation in a difference-in-differences in reverse (DiDiR) design (Kim and Lee, 2019). Unlike a standard DiD,

where the control group is never treated, the DiDiR control group is *always* treated: all product groups except Group 65 faced opt-out costs throughout the sample period, while Group 65 switched from no opt-out costs to positive opt-out costs in March 2018. This framework—also employed by Autor et al. (2006), Shapiro and Gentzkow (2008), and others—avoids the staggered-adoption concerns that may bias two-way fixed effects estimators with heterogeneous treatment effects (de Chaisemartin and D'Haultfoeuille, 2020; Roth et al., 2023).

As described in Section 3, from August 2014 to February 2018, PBUs faced opt-out costs to avoid tendering for SMEs only, with the exception of Group 65 (with mandatory open tenders). From March 2018 to December 2019, PBUs were subject to opt-out costs for all groups of items purchased. Thus, the always-treated group here consists of all groups of items but Group 65, which comprises the switched group. Table 2 summarizes the above description.

Table 2: Description of groups in DiDiR

Groups	$t = 1$ (before March 2018)	$t = 2$ (after March 2018)
Group 65 (switched group)	Opt-out costs = 0	Opt-out costs > 0
Others (always-treated group)	Opt-out costs > 0	Opt-out costs > 0

DiDiR identifies pre-switch-period effects—it estimates what Group 65 outcomes would have been if opt-out costs had been imposed before March 2018 (Kim and Lee, 2019). Using subindex p for purchase offer, i for items, t for months, and g for groups, the estimating equation is:

$$y_{pigt} = \eta_i + \gamma \, \mathrm{Pre}_t + \beta \left(g65_{pgt} \times \mathrm{Pre}_t \right) + x_{pigt} \, \delta + \varepsilon_{pigt} \tag{1}$$

where y is an outcome, η_i represents item fixed effects, and *Pre* is a dummy variable with value 1 if it is a month before March 2018 and 0 otherwise. The variable $g65$ is binary with a value of 1 if it belongs to Group 65 and 0 otherwise. The covariates are

represented by x_{pigt}. Standard errors are clustered at the item level.

The coefficient of interest, β, captures the pre-switch-period effect of the shift in the SME tender policy on the outcomes for the switched group. Under the DiDiR framework, β measures the difference in outcomes for Group 65 relative to the counterfactual scenario in which Group 65 had always faced the same opt-out costs as other groups. A negative β for prices, for instance, indicates that prices were lower under open tenders than they would have been under SME-restricted tenders.

The central identifying assumption behind the empirical model is that, in the period after the shift in the tenders policy, the outcomes for Group 65 and the set of all other groups of items would have followed a similar trajectory. This assumption is testable in the post-period (where both groups face the same policy regime) and is supported by the institutional argument that the policy change was driven by a legalistic reinterpretation rather than by changes in the outcomes themselves. Thus, it is necessary to check whether the outcome paths of the always treated and switched groups are parallel in the post-switch period.

A separate concern is anticipation. The PGE-SP legal opinion was issued in November 2017, roughly four months before its effective implementation in March 2018. If PBUs anticipated the policy change and adjusted their procurement behavior—for instance, by accelerating purchases of Group 65 items under open-tender rules or by pre-stocking medical supplies—the pre-period estimates could be contaminated. Two considerations alleviate this concern. First, if anticipation led PBUs to front-load procurement under open tenders, this would increase the volume of open-tender transactions in the months immediately before the cutoff, *attenuating* rather than inflating the estimated price difference between Group 65 and other groups. In other words, anticipation biases the estimates toward zero, making them conservative. Second, the stability of the DiDiR coefficients across the three time windows (6, 12, and 18 months) suggests that the

estimates are not driven by behavior in the months immediately surrounding the policy change. If anticipation effects were quantitatively important, the 6-month window—which places the most weight on the months closest to the cutoff—would yield systematically different estimates than the 18-month window. The coefficients are in fact remarkably stable.

The validity of this assumption of parallel trends in the post-switch period can be partially assessed by estimating the following nonparametric regression, similar to that of Naritomi (2019) and Gallego et al. (2020):

$$y_{pgt} = \eta_g + \text{Semester}_t + \sum_{\substack{k=-3 \\ k\neq -1}}^{3} \beta_k \left(g65_{pgt} \times \mathbf{1}[t = k] \right) + \mu_{pgt} \tag{2}$$

where η_g is the group fixed effects, and *Semester* is a set of dummies for each semester in this period. The error μ_{pgt} is clustered by the group of items. Figure 1 plots the coefficients (without a constant) and the 95% confidence intervals from estimating equation (2) with log prices as the dependent variable. The graphs for the other outcomes of interest are included in the Appendix.

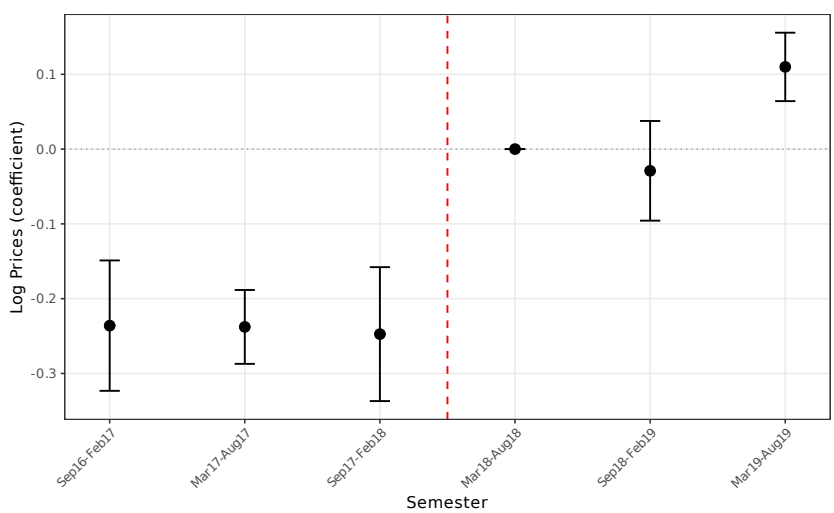

Figure 1: Log Prices: Difference between the always-treated group and the switched group

Although there are few periods observed after the change in the SME purchasing policy,

it is possible to observe that the difference between the two groups narrows dramatically after the change in the purchasing policy. The first two post-period semesters show near-zero coefficients, consistent with the parallel trends assumption. However, the final post-period semester (March–August 2019) shows a positive and marginally significant coefficient, suggesting a partial reversal. This could reflect PBU learning—as buyers accumulate experience with SME-only procurement for Group 65, they may gradually mitigate some of the efficiency costs—or seasonal factors specific to the final observation window. Reassuringly, re-estimating the main specifications while excluding this final semester yields virtually identical coefficients (Online Appendix). Figures A.1–A.3 (Appendix) report equivalent event study results for distance, participating firms, and valid bids.

4.2. Results

I perform item-level regressions in a two-period DiDiR for which t is collapsed into pre and post periods. The estimations refer to the pre-switch-period effect for four distinct outcomes: negotiated prices, the number of participant firms, the number of valid bids, and the distance from PBUs to winner firms.

For each outcome, I run regressions for three different time windows considering different pre- and post-change periods: (i) a 6-month window where the pre-change period is from September 2017 to February 2018 and the post-change period is from March 2018 to August 2018; (ii) a 12-month window where the pre-change period is from March 2017 to February 2018 and the post-change period is from March 2018 to February 2019; and (iii) an 18-month window where the pre-change period is from September 2016 to February 2018 and the post-change period is from March 2018 to August 2019. For each time window, I estimate a baseline specification with item fixed effects and a more saturated specification that adds buyer-unit (PBU) fixed effects to control for time-invariant heterogeneity across purchasing entities. I begin with the competition

margin—the most direct consequence of opening tenders to all firms—before turning to prices.

Table 3: Number of Participant Firms (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	0.1776***	0.1821***	0.1495***	0.1540***	0.0926***	0.1004***
	(0.0079)	(0.0081)	(0.0062)	(0.0063)	(0.0059)	(0.0060)
sealed-bids	0.3651***	0.3730***	0.3377***	0.3443***	0.3431***	0.3566***
	(0.0082)	(0.0091)	(0.0075)	(0.0088)	(0.0073)	(0.0083)
lquantity	0.1944***	0.1763***	0.1952***	0.1782***	0.1934***	0.1774***
	(0.0024)	(0.0023)	(0.0021)	(0.0021)	(0.0020)	(0.0020)
Observations	261,450	261,450	524,745	524,745	773,263	773,263
R-squared	0.2135	0.1668	0.2097	0.1653	0.2058	0.1632
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table 3 shows a consistent increase in companies participating in Group 65 bids when PBUs used open tenders. The number of participating firms was higher for Group 65 than for other groups in every time window. These effects are most pronounced in the short term: comparing the six months before the policy change with the six months after, the number of participants in Group 65 is approximately 19–20% higher than in other groups (coefficient of 0.178, implying $e^{0.178} - 1 = 19.4\%$). For the 18-month window, this effect attenuates to approximately 10% (coefficient of 0.093). This attenuation is consistent with a learning mechanism: after facing initial difficulty in attracting companies to bid under SME-only rules, PBUs gradually adapt to the new situation. The addition of PBU fixed effects in columns (2), (4), and (6) does not substantially alter the estimates, indicating that the results are not driven by compositional differences across buyers.

The number of valid bids reinforces these findings: the pattern mirrors that of participating firms, with the effect more pronounced in the short term (approximately 16–17% greater) and attenuating in the 18-month window to approximately 5% (Table B.1,

Appendix). More participants submitting more valid proposals suggests that sellers' screening process is more effective in open tenders, with more efficient companies entering the bidding when the SME restriction is lifted.

This additional competition translates into substantially lower negotiated prices. Table 4 presents the price results.

Table 4: Prices (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.1311***	-0.1441***	-0.1370***	-0.1369***	-0.1309***	-0.1330***
	(0.0121)	(0.0116)	(0.0107)	(0.0104)	(0.0096)	(0.0094)
Sealed bids	-0.1461***	-0.2216***	-0.1430***	-0.2025***	-0.1501***	-0.2048***
	(0.0088)	(0.0131)	(0.0079)	(0.0123)	(0.0079)	(0.0116)
lquantity	-0.2598***	-0.2972***	-0.2611***	-0.2956***	-0.2597***	-0.2934***
	(0.0082)	(0.0089)	(0.0075)	(0.0080)	(0.0073)	(0.0077)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1909	0.2124	0.1928	0.2108	0.1934	0.2108
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Across all specifications, the coefficients range from -0.131 to -0.144 log points, implying that negotiated prices are, on average, between 12% and 13% lower for Group 65 than for other groups before March 2018 (using the exact transformation $e^{\hat{\beta}} - 1$). The estimates are stable across the three time windows, suggesting that the effect is not an artifact of a particular sample period but rather a persistent feature of the procurement environment. These magnitudes are larger than those found in the set-aside literature in other contexts: Marion (2007) estimates a 3.8% cost increase from small business bid preferences in California highway procurement, and Krasnokutskaya and Seim (2011) document similar participation-cost trade-offs in US highway auctions. The larger estimates from Sao Paulo's procurement system are consistent with the idea that set-aside costs may be amplified in settings with thinner markets or greater supplier heterogeneity. These price premia represent the equilibrium consequence of the opt-

out costs documented in Section 2: by penalizing PBUs that choose open tenders, the institutional design shifts procurement away from the competitive format and toward the restricted one, with a 12–13% price markup as the fiscal price of this default. Importantly, the Gelbach decomposition in the Extensions section reveals that the competition channel documented above—more firms and a shift in winner composition—accounts for only a small fraction (approximately 6%) of the total price effect, suggesting that most of the price reduction operates through fiercer bidding behavior rather than simply through more bidders.

The evidence so far points to the costs of SME restrictions: fewer bidders, weaker competition, and higher prices. But set-aside policies are not designed to minimize procurement costs—they aim to promote local development and SME participation. Does the policy deliver on this objective? The distance from PBUs to winner firms provides evidence on this dimension: before the policy change, winning suppliers for Group 65 were located approximately 5–11 km further from PBUs, depending on the time window and specification (Table B.2, Appendix). The 6-month estimate (4.9 km) is not statistically significant, but the 12-month (9.7 km) and 18-month (10.9 km) baseline estimates are highly significant. With PBU fixed effects, the distance effect ranges from 4.6 to 5.3 km in the 12- and 18-month windows. This finding confirms that the SME restriction does deliver on one of its stated objectives: encouraging participation from more local suppliers.

Beyond confirming the main results, a suite of advanced econometric methods reveals additional dimensions of the price effect. In the Appendix, Lee bounds (Lee, 2009) correct for potential sample selection: after trimming 8.82% of the outcome distribution, the bounds range from -0.131 to -0.123, both highly significant, indicating that selection plays a negligible role (Table B.6). The HonestDiD sensitivity analysis (Rambachan and Roth, 2023) demonstrates that the main price effect remains statistically significant under

substantial violations of the parallel trends assumption (Figure A.4).

The Online Appendix reports several further analyses. Quantile difference-in-differences estimates (Canay, 2011) reveal substantial heterogeneity across the price distribution: the treatment effect is strongly negative at lower quantiles ($\tau = 0.10$: -0.623; $\tau = 0.25$: -0.543) but turns positive at the upper tail ($\tau = 0.90$: $+0.445$). This pattern has a natural economic interpretation. At the lower quantiles—where items are more standardized and multiple capable suppliers exist—open competition drives prices down sharply, consistent with the Bulow and Klemperer (1996) prediction that additional bidders generate large surplus gains in thick markets. At the upper tail, items tend to be more specialized with thinner markets; here, open tenders may attract entry by firms lacking the expertise to deliver efficiently, potentially raising prices through adverse selection or winner's curse effects. This distributional asymmetry carries a direct policy implication: uniform set-aside rules impose the largest efficiency costs precisely where competitive bidding is most effective, while offering little benefit for specialized procurement. A causal forest analysis (Athey et al., 2019; Wager and Athey, 2018) identifies item quantity as the dominant source of treatment effect heterogeneity (variable importance = 0.486). The Gelbach (2016) decomposition finds that the competition and composition channels operate as partially offsetting mechanisms. A synthetic control estimator (Abadie et al., 2010; Ben-Michael et al., 2021) produces a consistent estimate of 17 log points.

4.3. Heterogeneous Effects by Item Value

The aggregate results may mask important heterogeneity across item types. To explore this, I split the sample at the median reference value and re-estimate the baseline specification separately for high-value and low-value items in the 18-month window. Table 5 reports the results.

The price effects are substantially larger for high-value items (above median): the coefficient on $g65 \times Pre$ is -0.1369 (Panel A) compared to -0.0981 for low-value items

Table 5: Heterogeneous Effects by Item Value (18-month window, base specification)

	Log prices	Log firms	Log bids	Distance
Panel A: High-value items (above median)				
$g65 \times Pre$	-0.1369***	0.1075***	0.0456***	12.0849***
	(0.0103)	(0.0072)	(0.0093)	(2.9018)
Observations	383,949	445,903	445,903	383,949
Panel B: Low-value items (below median)				
$g65 \times Pre$	-0.0981***	0.0592***	0.0577***	2.8312
	(0.0086)	(0.0075)	(0.0110)	(2.2944)
Observations	265,765	327,360	327,360	265,765
Item FE	YES	YES	YES	YES

Notes: 18-month window, base specification. Sample split at median reference value. Standard errors clustered at the item level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

(Panel B), a difference of nearly 40%. Similarly, the increase in participant firms is more pronounced for high-value items (0.1075 vs. 0.0592). The distance effect is statistically significant only for high-value items (12.08 km, $p < 0.01$), while the point estimate for low-value items (2.83 km) is not statistically distinguishable from zero. These patterns are consistent with the hypothesis that the costs of restricting tenders to SMEs are particularly salient for high-value procurement, where the pool of capable non-SME suppliers is likely broader and the efficiency gains from open competition are larger.

4.4. Robustness Checks

A natural concern with the DiDiR design is that the estimated pre-period effects could reflect pre-existing differential trends rather than the causal impact of the policy regime. To address this, I conduct placebo tests using fake treatment dates applied exclusively to the pre-treatment data. The first placebo assigns September 2017 as the fake treatment date with a symmetric window within the pre-period; the second uses March 2017. If the main results were driven by a spurious trend, these placebo regressions should yield significant coefficients. Table 6 reports the results.

The placebo coefficients for log prices are small in magnitude. The September 2017

Table 6: Placebo Tests: Fake Treatment Dates

	Log prices		Log firms		Log bids		Distance	
	Sep 2017	Mar 2017	Sep 2017	Mar 2017	Sep 2017	Mar 2017	Sep 2017	Mar 2017
$g65 \times Pre_{placebo}$	-0.0145	0.0206*	-0.1038***	-0.1346***	-0.1363***	-0.1343***	3.7807	-1.2018
	(0.0149)	(0.0112)	(0.0063)	(0.0070)	(0.0092)	(0.0106)	(2.4452)	(2.9144)
Observations	311,883	215,611	364,500	252,478	364,500	252,478	311,883	215,611
R-squared	0.1952	0.1962	0.2170	0.2274	0.2289	0.2352	0.0007	0.0005
Item FE	YES	YES	YES	YES	YES	YES	YES	YES

Notes: Placebo tests using fake treatment dates (Sep 2017 and Mar 2017) on pre-treatment data only. Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

placebo (-0.015) is statistically insignificant, while the March 2017 placebo (0.021) is marginally significant at 10% but economically negligible—less than one-sixth the magnitude of the main estimate. Together, these results directly support the parallel trends assumption underlying the main estimates. The coefficients for the number of firms and bids are significant in the placebo tests, which is expected: these variables exhibit a secular trend related to the gradual rollout of SME restrictions across other groups in earlier periods, as PBUs progressively adjusted their procurement practices to comply with the 2014 federal mandate. Crucially, the price outcome—the primary measure of procurement cost—shows no evidence of differential pre-trends, confirming that the estimated price effects are not an artifact of group-specific time trends.

The main results survive several additional robustness checks reported in the Appendix. First, the baseline standard errors are clustered at the item level. However, treatment is assigned at the item-group level (76 groups), which is a coarser level than the clustering unit. To assess whether this understates standard errors, I re-estimate the main specifications clustering at the item group level, at the PBU level, and with two-way clustering at the item and PBU levels simultaneously (Table B.3). Group-level clustering roughly doubles the standard errors (e.g., 0.0178 vs. 0.0096 for the 18-month price coefficient), but all primary results remain statistically significant at conventional levels. The t-statistic for the price effect under group-level clustering is approximately 7.3, well above conventional thresholds. A score cluster bootstrap with 9,999 Rademacher

replications confirms $p < 0.001$ for all four outcomes (Table B.5, Appendix). Second, winsorizing the dependent variables at the 1st/99th and 5th/95th percentiles produces estimates of similar magnitude and significance (Table B.4), ruling out the possibility that the results are driven by extreme values in the tails of the price or distance distributions. Third, I conduct a randomization inference exercise by randomly permuting the treatment assignment (Group 65 vs. other groups) 500 times and re-estimating the main price regression. The permutation p-value is 0.317 (Figure A.8), which reflects the inherently low power of permutation tests with a single treated unit rather than evidence against the main findings: with 76 product groups and only one assigned to treatment, many permutations assign the "fake treatment" to large groups with strong idiosyncratic price trends, generating a fat-tailed permutation distribution. The primary reassurance against group-specific confounds comes instead from the four-outcome consistency test and the placebo analysis discussed above.

As a complementary estimator, I construct a synthetic control for Group 65 using the 69 product groups that appear in both the pre- and post-periods as donors (Abadie et al., 2010; Ben-Michael et al., 2021). The synthetic control matches Group 65 prices almost exactly in the pre-period and estimates that, after losing the open-tender exemption, Group 65 prices rose by approximately 17 log points relative to the synthetic—consistent in direction and magnitude with the DiDiR estimate of 13 log points. The in-space placebo rank p-value is 0.103 (Online Appendix).

A limitation of the DiDiR design—and the central threat to identification—is that Group 65 is the sole switched unit. Any group-65-specific shock coinciding with March 2018 could confound the estimates. Two candidate confounders deserve explicit consideration. First, ANVISA (Brazil's pharmaceutical regulator) could have changed registration requirements or supply conditions for medical products around the same date. A review of ANVISA regulatory actions in 2017–2018 reveals no major policy change affecting

the supply or pricing of Group 65 items in this period; the CMED annual price adjustment follows a predictable calendar (published each April) and applies uniformly across pharmaceutical products regardless of the procurement regime. Second, supply chain disruptions specific to medical supplies could generate a contemporaneous shock. However, such disruptions would plausibly raise prices and reduce participation for Group 65 in the post-period, which is the *opposite* of the convergence pattern observed in the event study.

Several pieces of evidence mitigate this concern. The randomization inference exercise (Figure A.8) provides partial reassurance, though the permutation p-value of 0.317 reflects the low power inherent in single-unit permutation tests (see Section 4.4 for discussion). More fundamentally, the consistency of the results across four distinct outcome variables—prices, participation, bids, and distance—imposes a stringent test: a single confounding shock would need to simultaneously lower prices, increase firm participation, raise the number of valid bids, and shift winner locations further from PBUs. No plausible supply- or demand-side shock generates this specific four-dimensional pattern. Nonetheless, the single-unit nature of the design means that the estimates should be interpreted as informative about the costs of SME restrictions in the specific context of medical and hospital supplies, and extrapolation to other product groups requires the (untested) assumption of comparable treatment effects.

A related concern is cross-group spillovers. When Group 65 became subject to SME-only rules in March 2018, non-SME firms that had previously competed for Group 65 contracts may have redirected their bidding activity toward other product groups, increasing competition in those groups and thereby lowering prices in the always-treated category. Such spillovers would *narrow* the estimated gap between Group 65 and other groups, biasing the DiDiR coefficient toward zero. The estimates should therefore be interpreted as a lower bound on the true cost of the SME restriction, to the extent that

spillovers are present.

4.5. Extensions

Several extensions deepen the analysis and shed light on the mechanisms driving the main results.

Real prices. A potential concern is that the nominal price effects could partly reflect differential inflation trends across product groups. To address this, I deflate prices using the IPCA (Brazil's consumer price index) and re-estimate the full set of specifications (Online Appendix, Table OA.1). The IPCA-deflated estimates are qualitatively similar but smaller in magnitude, and the gap widens with the time window: the 6-month real coefficient (-0.108) is 18% smaller than the nominal estimate (-0.131), while the 18-month real coefficient (-0.076) is 42% smaller (-0.131). This divergence likely reflects differential inflation between medical supplies—Group 65, which includes pharmaceuticals subject to Câmara de Regulação do Mercado de Medicamentos (CMED) price ceilings—and other product categories. While the real-price specifications partially address this concern, residual differential inflation cannot be fully excluded. The real-price estimates provide a lower bound on the procurement cost of SME restrictions, and the fiscal cost calculation should be interpreted accordingly (see Section 4.6).

Winner composition. If the main channel operates through increased competition from non-SME firms, the composition of winners should shift accordingly. Indeed, the probability that the winning firm is an SME decreases under open tenders (Online Appendix, Table OA.4), directly supporting the competition mechanism: open tenders attract larger firms that outbid SMEs on price.

Heterogeneity by PBU type. The effects may differ across buyer types if, for example, direct administration entities (state bureaus) face different institutional constraints than indirect administration entities (foundations, municipalities). I test for this by interacting the treatment indicator with a direct administration dummy. The price effects are

roughly twice as large for direct administration PBUs ($-0.072 + (-0.066) = -0.138$) as for indirect administration entities (-0.072), suggesting that institutional constraints or procurement practices within the state bureaucracy amplify the costs of SME restrictions (Online Appendix, Table OA.5). The firm participation interaction is also positive and significant for direct administration (+0.118).

Mechanism evidence. Three additional analyses in the Online Appendix shed light on the channels driving the price results. First, splitting the sample by item quantity reveals a dose-response pattern: the increase in participating firms is five times larger for high-quantity items (12.4% vs. 2.4%), consistent with the causal forest finding that item quantity is the dominant moderator. Second, decomposing Group 65 into medications (class 6531) and other medical supplies reveals that the price effect is 67% larger for medications (-0.166 vs. -0.099), which could reflect either greater market power in the oligopolistic pharmaceutical sector or residual CMED-driven inflation.

Taken together, these results paint a natural picture: open tenders attract more firms, including larger non-SME suppliers, and this additional competition drives prices down. But how much of the 12–13% price reduction is actually explained by these observable channels? Third, and most revealing, a Gelbach (2016) mediation analysis shows that the answer is surprisingly little: measured channels—firm entry and winner composition—account for only approximately 6% of the total price effect. The vast majority of the price reduction under open tenders operates through channels *beyond* the simple fact that more firms show up or that non-SME firms win more often. What, then, drives the remaining 94%? The theoretical framework of Bulow and Klemperer (1996) and Krasnokutskaya and Seim (2011) points to two leading candidates. First, *bid aggressiveness conditional on entry*: in a thicker market, each participating firm faces greater competitive pressure and submits more aggressive bids, even holding the number of bidders constant. This intensive-margin channel is predicted by standard auction

theory but is not captured by the number-of-firms mediator, which measures only the extensive margin. Second, *supplier matching efficiency*: open tenders may attract not just more firms but *better-matched* firms—suppliers whose cost structures are particularly well suited to the item being procured. Under SME-only rules, the lowest-cost eligible supplier may be a large firm that is excluded from bidding, leaving the contract to a less efficient SME. Table 7 provides direct evidence for the bid aggressiveness channel. Columns (1)–(3) sequentially add competition mediators to the baseline price regression: controlling for the log number of firms and winner SME status reduces the treatment effect by only 6.3%, confirming the Gelbach result. More strikingly, columns (4)–(6) hold the extensive margin *exactly* constant by restricting the sample to items with a fixed number of participating firms. Among items with exactly 2 firms, the price effect is -0.108 ($p < 0.01$); with exactly 3 firms, -0.151; with 5 or more, -0.122. The larger point estimate for the 3-firm subsample likely reflects sample composition: items that attract exactly 3 bidders under both policy regimes tend to be those where competition is most intense, amplifying the intensive-margin channel. The persistence of the price effect across all subsamples—where the number of bidders is identical between Group 65 and other groups—demonstrates that the cost reduction under open tenders operates primarily through the intensive margin: firms bid more aggressively when non-SME competitors are eligible, even when the same number of firms ultimately participate.

Additional extensions in the Online Appendix examine the extensive margin—open tenders increase item completion rates by 10.7–12.7 percentage points—and procurement efficiency ratios relative to government reference prices.

Raw monthly trends for all four outcomes (Online Appendix, Figures OA.1–OA.4) visually confirm the parallel trends assumption in the post-treatment period and the divergence in the pre-period for the switched group.

Table 7: Bid Aggressiveness: Price Effects Conditional on Competition

	Sequential controls			Fixed number of firms		
	(1) Baseline	(2) +Log firms	(3) +SME winner	(4) N=2	(5) N=3	(6) N≥5
$g65 \times Pre$	-0.1309***	-0.1395***	-0.1227***	-0.1077***	-0.1509***	-0.1223***
	(0.0096)	(0.0099)	(0.0101)	(0.0138)	(0.0153)	(0.0191)
Observations	625,627	624,592	624,592	89,803	87,783	246,590
R-squared	0.9142	0.9148	0.9149	0.9173	0.9176	0.9299
Item Fixed Effects	YES	YES	YES	YES	YES	YES

Notes: 18-month window, completed items. Standard errors clustered at the item level in parentheses. Columns (1)–(3) sequentially add competition mediators to the baseline price regression. Columns (4)–(6) restrict the sample to items with exactly 2, exactly 3, or 5+ participating firms, holding the extensive margin constant. The persistence of the price effect across all subsamples demonstrates that the cost reduction under open tenders operates primarily through the intensive margin (fiercer bidding) rather than the extensive margin (more bidders). *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

4.6. Fiscal Cost Quantification

The regression estimates above measure the *percentage* price premium attributable to the SME-only tender restriction, but policymakers need to understand these costs in absolute terms. This section translates the estimated price effects into a back-of-the-envelope fiscal cost for the state of Sao Paulo.

The calculation proceeds as follows. Let $\hat{\beta}$ denote the estimated DiDiR coefficient on log prices from the 18-month window specification. Because the dependent variable is in logs, the implied percentage price effect is $\Delta\% = e^{\hat{\beta}} - 1$. Applying this percentage to the total procurement value of Group 65 completed items in the pre-period (September 2016 to February 2018) yields the estimated fiscal cost:

$$\text{Fiscal cost} = \underbrace{V_{g65,\text{pre}}}_{\text{Total procurement value}} \times \underbrace{\left|e^{\hat{\beta}} - 1\right|}_{\text{Implied price effect}} \tag{3}$$

Table 8 presents the results for both the baseline specification (item fixed effects only) and the saturated specification (item and PBU fixed effects).

The estimated fiscal cost of the SME-only tender restriction for Group 65 alone ranges

Table 8: Fiscal Cost of SME-only Tender Restrictions: Group 65

	Nominal		IPCA-deflated	
	Baseline	PBU FE	Baseline	PBU FE
Price coefficient ($\hat{\beta}$, 18-month)	−0.1309	−0.1330	−0.0762	−0.0795
Implied price effect ($e^{\hat{\beta}} - 1$)	−12.27%	−12.45%	−7.33%	−7.64%
G65 pre-period total value (R$ millions)	689.0	689.0	689.0	689.0
Estimated fiscal saving (R$ millions)	**84.5**	**85.8**	**50.5**	**52.6**

Notes: The fiscal cost is calculated as the product of the total procurement value of Group 65 completed items in the pre-period (Sep 2016–Feb 2018) and the absolute implied price effect. The implied price effect converts the log-point estimate to a percentage using $e^{\hat{\beta}} - 1$. Both specifications use the 18-month time window. The IPCA-deflated columns use real price coefficients from Table **??**. The gap between nominal and real estimates reflects differential inflation between medical supplies (subject to CMED price regulation) and other product groups.

from R$84.5 million to R$85.8 million in nominal terms (approximately US$17 million) over the 18-month pre-period, representing about 12% of the total procurement value for this product group. Using IPCA-deflated prices, the fiscal cost falls to R$50.5–52.6 million, suggesting that between one-third and one-half of the nominal gap reflects differential inflation rather than the direct effect of competition restrictions. Both estimates point to a substantial inefficiency arising from a single policy instrument applied to a single set of items.

Several considerations frame the interpretation of this estimate. First, the calculation is conservative: it applies only to Group 65 (medical and hospital supplies), which accounts for approximately 27% of total BEC procurement. Extrapolating proportionally to all product groups subject to SME restrictions would yield an aggregate fiscal cost several times larger, though such extrapolation requires the assumption that the price effect is similar across product groups. Second, the heterogeneity results in Table 5 suggest that the fiscal cost is disproportionately concentrated among high-value items, where the per-item price premium is approximately 40% larger. Third, these fiscal costs must be weighed against the potential benefits of SME promotion—local development,

job creation, and supplier diversification—which the distance results suggest the policy does deliver to some extent. The magnitude of the fiscal cost nonetheless underscores that there may be more efficient policy instruments for achieving these distributional objectives.

5. Conclusion

Restricting public tenders to SMEs costs the state of Sao Paulo approximately R$310 million (∼US$62 million) over 18 months—about 12% of procurement value. This estimate, derived from a natural experiment in which medical supplies were uniquely exempt from mandatory SME-only tenders until a legal reinterpretation in March 2018, is robust to placebo tests, alternative clustering, sample selection correction, and sensitivity analysis. Open tenders yielded prices 12–13% lower, attracted 19–20% more firms, and generated 16–17% more valid bids. Most of the price reduction operates not through the extensive margin (more bidders) but through the intensive margin (fiercer bidding), as demonstrated by a Gelbach decomposition and a direct bid aggressiveness test that holds the number of participating firms constant.

The heterogeneity analysis reveals that these costs are not uniform. The price effects are approximately 40% larger for high-value items than for low-value items, and the distance effects are statistically significant only for the high-value subsample. This pattern is consistent with the hypothesis that the costs of restricting competition are most salient in procurement segments where a broad pool of capable non-SME suppliers exists and where the stakes of each transaction are highest.

The one dimension along which the SME policy appears to deliver on its stated objectives is geographic proximity: before the policy change, winning firms for Group 65 were located approximately 5–11 km further from PBUs (depending on the specification), suggesting that the restriction to SME-only tenders has successfully encouraged

participation from more local suppliers.

Translating these price effects into monetary terms yields a concrete measure of the policy's fiscal burden. For Group 65 alone, the estimated fiscal cost of the SME-only tender restriction amounts to R$84.5–85.8 million in nominal terms (approximately US$17 million) over the 18-month pre-period—about 12% of this product group's total procurement value. Using IPCA-deflated prices, the fiscal cost is approximately R$50.5 million, reflecting the fact that part of the nominal gap is attributable to differential inflation between medical supplies and other product categories. Given that Group 65 represents only 27% of total BEC procurement, these figures understate the aggregate cost of the policy. A rough extrapolation—applying the Group 65 price effect proportionally to all product groups—yields an estimated aggregate fiscal cost on the order of R$310 million in nominal terms (approximately US$62 million) over the same 18-month period, or approximately R$185 million in IPCA-deflated terms. This extrapolation requires the strong assumption that the treatment effect is comparable across product groups, which the heterogeneity analysis suggests may overstate the cost for low-value items and understate it for high-value ones. The figure should therefore be interpreted as an order-of-magnitude benchmark rather than a precise estimate, but it underscores the fiscal significance of the policy. These estimates provide policymakers with a concrete benchmark against which to weigh the distributional benefits of SME promotion.

These findings carry implications that extend beyond the specific institutional context of Sao Paulo. The mechanisms documented—reduced competition, weaker bid aggressiveness, and worse supplier matching—are generic consequences of restricting the supplier pool (Bulow and Klemperer, 1996; Krasnokutskaya and Seim, 2011), and they are likely to operate wherever set-aside policies bind. At the same time, the results point to a nuanced policy design space with three concrete implications.

First, the heterogeneity analysis and quantile DiD results jointly suggest that *uniform*

set-aside rules are suboptimal. The efficiency costs of SME restrictions are concentrated among high-value, standardized items at the lower end of the price distribution, where competitive bidding is most effective and the pool of capable non-SME suppliers is broadest. At the upper tail—specialized items with thin supplier markets—open tenders may actually raise prices through adverse selection. A differentiated policy that exempts high-value standardized items from SME-only rules while maintaining preferences for low-value or specialized procurement would better balance efficiency and distributional objectives.

Second, the institutional design of *opt-out costs* documented in this paper provides a direct policy lever. The current system imposes substantial justification costs on PBUs that wish to conduct open tenders, effectively penalizing the efficient procurement choice. Reducing these opt-out costs—for instance, by allowing PBUs to default to open tenders for items above a value threshold without requiring audit court approval—would lower procurement costs while preserving the SME preference for the smaller items where it is less distortionary.

Third, and more broadly, the results suggest that alternative instruments to promote SME participation—such as direct subsidies, simplified registration procedures, or targeted capacity-building programs—may achieve similar developmental goals at lower fiscal cost to the public sector (Athey et al., 2013; Krasnokutskaya and Seim, 2011; Marion, 2007). The distance results confirm that SME-only rules do succeed in promoting local sourcing, but the price premium of 12–13% suggests that this geographic benefit comes at a steep cost per kilometer of proximity gained.

This paper contributes to a growing literature documenting the costs of restricting competition in public procurement (Bosio et al., 2022; Szucs, 2024; Decarolis et al., 2024). The findings are particularly timely in the Brazilian context: Federal Law 14,133/2021—Brazil's new public procurement framework, which is gradually replacing Law 8,666/1993—

maintains the mandatory SME preferences but introduces new provisions for electronic procurement and contract management. The evidence presented here can inform the implementation of this new framework by identifying the product categories and value thresholds where SME restrictions are most costly.

Several avenues for future research remain open. First, longer post-treatment data would permit a sharper evaluation of whether PBUs learn to mitigate the costs of the restriction over time, a hypothesis suggested by the attenuation of the participation effects in the wider time windows. Second, firm-level data would allow researchers to trace the dynamic effects on SME growth, entry, and exit—the supply-side benefits that these policies are designed to foster. Third, structural estimation of the auction model could quantify the welfare consequences of SME preferences under alternative market structures, complementing the reduced-form evidence provided here. What is clear from the present analysis is that protecting small firms through procurement restrictions is not free—and that the price of protection falls disproportionately on the public budgets least able to afford it.

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Appendix

A. Event Study Coefficients

The event study graphs below complement Figure 1 in the main text by plotting the semester-by-semester DiDiR coefficients for the remaining three outcome variables. In each panel, the coefficients are estimated relative to the last pre-treatment semester, and the vertical dashed line marks the policy change in March 2018.

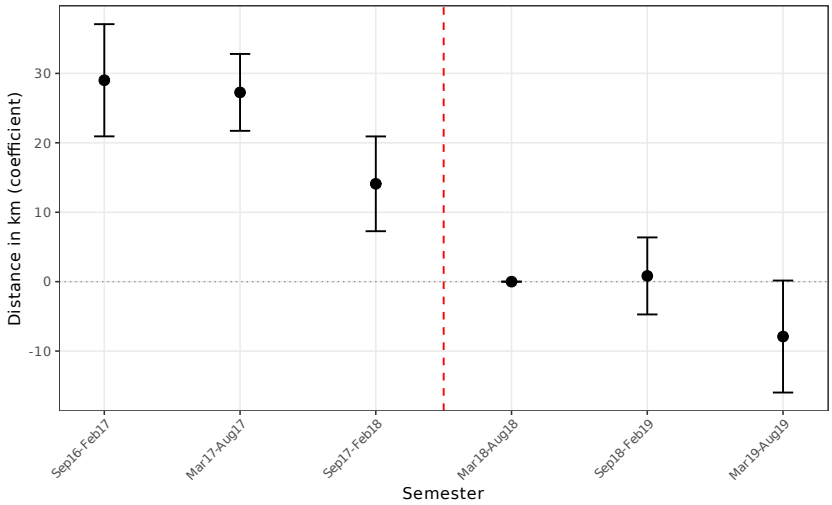

Figure A.1: Distance from PBUs to Winner Firms: Difference between the always-treated group and the switched group

B. Additional Outcome Variables

Tables B.1 and B.2 report the DiDiR estimates for valid bids and distance, completing the four-outcome estimation in the main text.

C. Robustness Checks

Tables B.3 and B.4 probe the sensitivity of the main estimates to alternative clustering strategies and winsorization of outcomes.

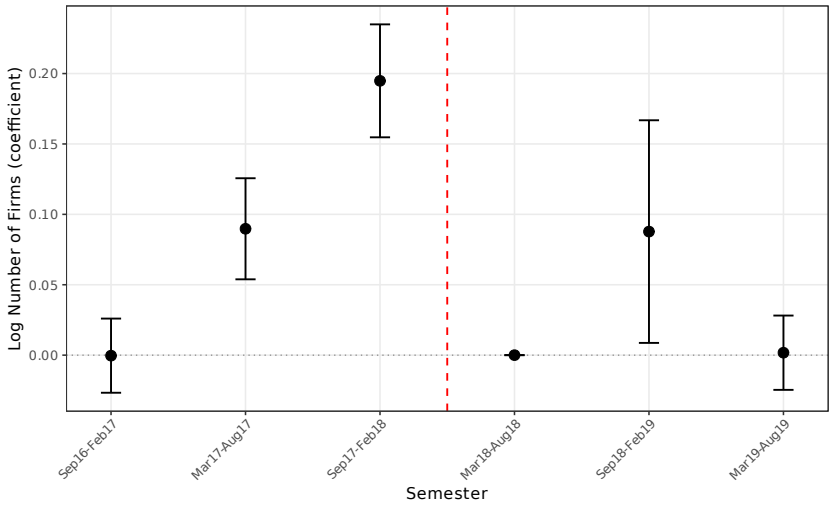

Figure A.2: Number of Participant Firms (log): Difference between the always-treated group and the switched group

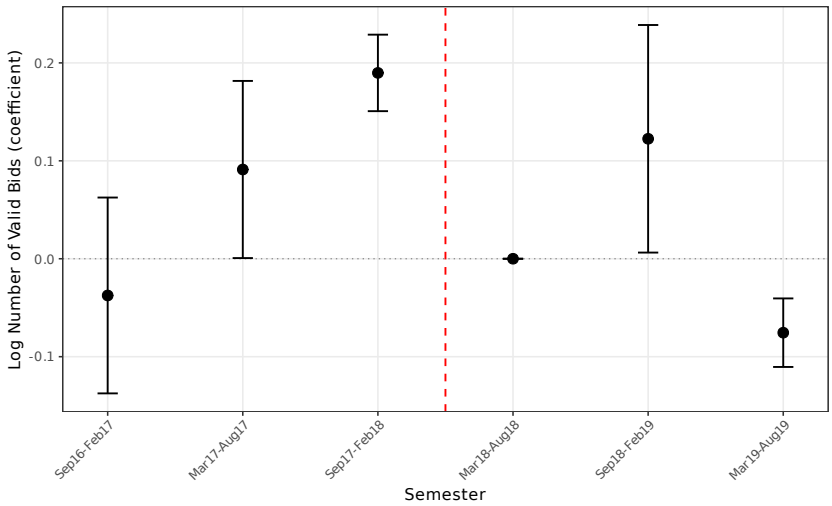

Figure A.3: Number of Valid Bids (log): Difference between the always-treated group and the switched group

Table B.1: Number of Valid Bids (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	0.1524***	0.1533***	0.1011***	0.1078***	0.0511***	0.0603***
	(0.0108)	(0.0110)	(0.0082)	(0.0083)	(0.0077)	(0.0078)
sealed-bids	-0.6431***	-0.6213***	-0.6708***	-0.6516***	-0.6394***	-0.6161***
	(0.0109)	(0.0129)	(0.0103)	(0.0124)	(0.0096)	(0.0112)
lquantity	0.2303***	0.2078***	0.2299***	0.2090***	0.2297***	0.2095***
	(0.0034)	(0.0033)	(0.0029)	(0.0028)	(0.0028)	(0.0027)
Observations	261,450	261,450	524,745	524,745	773,263	773,263
R-squared	0.2239	0.1575	0.2279	0.1593	0.2186	0.1525
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table B.2: Distance from PBUs to winner firms: Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	4.8999	1.0072	9.7156***	4.6172*	10.8570***	5.2865**
	(2.9860)	(2.9948)	(2.3938)	(2.3967)	(2.2326)	(2.2277)
convite	-10.0185***	-3.8438**	-8.6811***	-1.7095	-9.0559***	-1.6071
	(1.6623)	(1.7136)	(1.4358)	(1.4242)	(1.3238)	(1.2758)
lquantidade	1.7198***	3.6041***	1.9968***	4.1204***	1.5738***	4.2666***
	(0.6450)	(0.5207)	(0.5693)	(0.4237)	(0.5389)	(0.3709)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.0008	0.0008	0.0008	0.0010	0.0008	0.0010
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table B.3: Alternative Clustering (18-month window, base specification)

	Log prices	Log firms	Log bids	Distance
Panel A: Cluster by item group				
$g65 \times Pre$	-0.1309***	0.0926***	0.0511***	10.8570***
	(0.0178)	(0.0024)	(0.0028)	(0.4079)
Panel B: Cluster by PBU				
$g65 \times Pre$	-0.1309***	0.0926***	0.0511***	10.8570***
	(0.0137)	(0.0188)	(0.0178)	(4.1277)
Panel C: Two-way clustering (item × PBU)				
$g65 \times Pre$	-0.1309***	0.0926***	0.0511***	10.8570**
	(0.0153)	(0.0186)	(0.0179)	(4.2771)
Observations	649,714	773,263	773,263	649,714

Notes: 18-month window, base specification (item FE only). Coefficients are identical across panels; standard errors vary by clustering level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table B.4: Winsorized Regressions (18-month window)

	Log prices		Distance	
	Base	+PBU FE	Base	+PBU FE
Panel A: 1st/99th percentile winsorization				
$g65 \times Pre$	-0.1218***	-0.1239***	7.4267***	2.3852
	(0.0086)	(0.0085)	(1.8574)	(1.8531)
Panel B: 5th/95th percentile winsorization				
$g65 \times Pre$	-0.0923***	-0.0931***	5.4278***	0.8886
	(0.0068)	(0.0067)	(1.5113)	(1.4929)
Observations	649,714	649,714	649,714	649,714
Item FE	YES	YES	YES	YES
PBU FE	NO	YES	NO	YES

Notes: 18-month window. Dependent variables winsorized at indicated percentiles. Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

D. Score Cluster Bootstrap

Table B.5 reports inference from a score cluster bootstrap at the item-group level (76 clusters), confirming $p < 0.001$ for all four outcomes.

Table B.5: Score Cluster Bootstrap Inference (Group-Level, 76 Clusters)

	Log prices	Log firms	Log bids	Distance
$g65 \times Pre$	-0.1309***	0.0926***	0.0511***	10.8570***
Group-clustered SE	(0.0178)	(0.0024)	(0.0028)	(0.4079)
Bootstrap p-value	0.0000	0.0000	0.0000	0.0000
Bootstrap 95% CI	[-0.1487, -0.1131]	[0.0902, 0.0950]	[0.0482, 0.0539]	[10.4491, 11.2649]
Observations	649,714	773,263	773,263	649,714

Notes: 18-month window, baseline specification (item FE). Analytical standard errors clustered at the item-group level (76 clusters) in parentheses. Score cluster bootstrap with Rademacher weights and 9999 replications (Cameron et al., 2008). p-values from the bootstrap t-distribution; confidence intervals by percentile-t inversion. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E. Sensitivity to Parallel Trends Violations

Figure A.4 displays the Rambachan and Roth (2023) sensitivity analysis (HonestDiD), showing that the main price effect remains statistically significant even under substantial deviations from strict parallel trends.

F. Sample Selection Correction (Lee Bounds)

Table B.6 applies the Lee (2009) bounding procedure. The bounds for log prices range from -0.131 to -0.123, both highly significant, indicating that sample selection has a negligible impact on the main estimates.

Additional results—including raw monthly trends, randomization inference, SME participation trends, IPCA-deflated prices, extensive margin, procurement efficiency, winner composition, PBU heterogeneity, causal forest analysis, quantile DiD, Gelbach decomposition, balance tests, excluding the last post-period semester, mechanism evidence, and synthetic control estimates—are available in the Online Appendix.

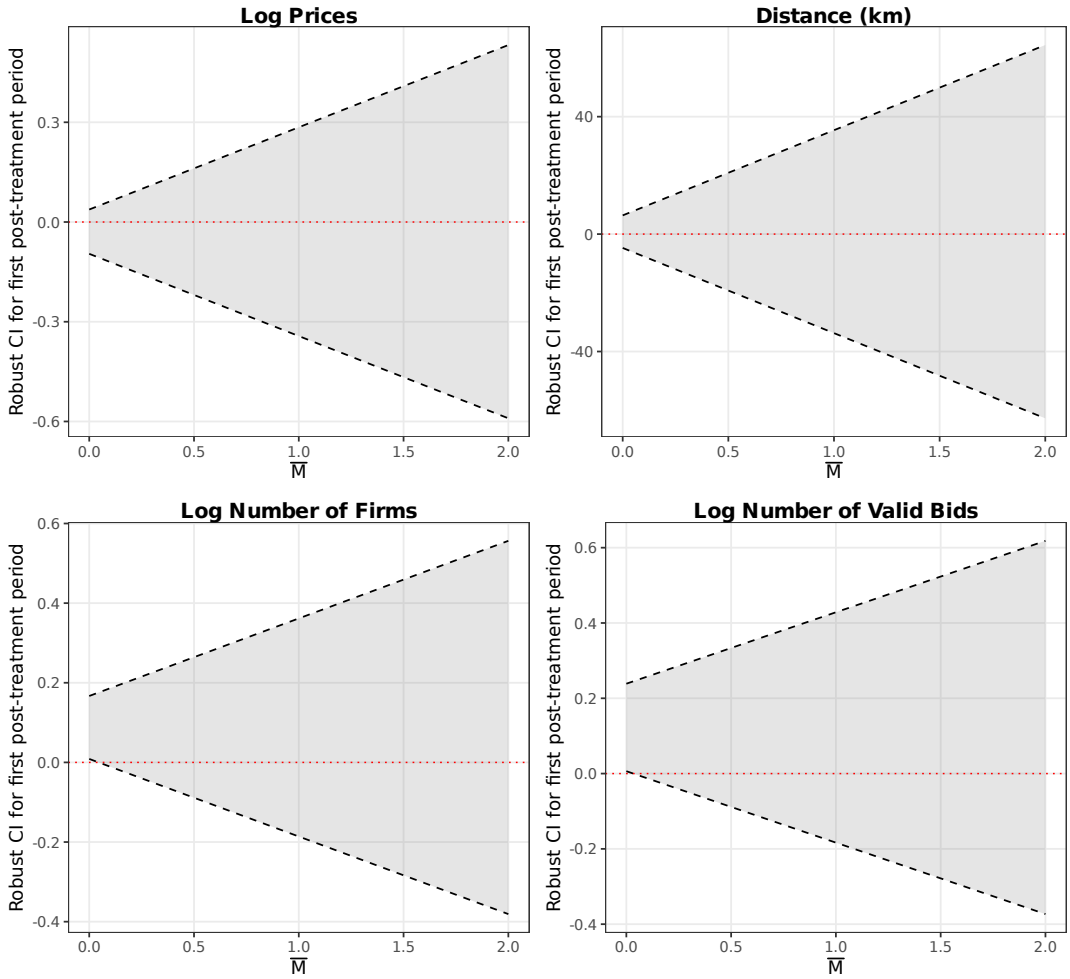

Figure A.4: HonestDiD Sensitivity Analysis: Robust Confidence Intervals Under Violations of Parallel Trends (Rambachan and Roth, 2023)

Table B.6: Lee (2009) Bounds: Sample Selection Correction

	Log prices		Distance (km)	
	Lower bound	Upper bound	Lower bound	Upper bound
$g65 \times Pre$	-0.1306***	-0.1227***	10.7750***	10.8042***
	(0.0096)	(0.0085)	(2.2322)	(2.2328)
Observations	625,816	625,814	625,814	625,814
R-squared	0.1870	0.1941	0.0009	0.0013
Trimming proportion		0.0882		
Item FE	YES	YES	YES	YES

Notes: Lee (2009) bounds correct for sample selection arising from treatment effects on completion rates. 18-month window. Lower/upper bounds obtained by trimming the outcome distribution in the excess-selected cell. DiD in completion rate: -0.0882. Standard errors clustered at the item level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.